WEB103 | Advanced Web Development

Advanced Web Development Summer 2026 (@ Section 1 | Wednesdays 5PM - 7PM PT)
Personal Member ID#: **150502**

Need help? Post on our [Class Slack Channel] or email us at [support@codepath.org]

Getting Started

Learning with AI ✨

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Course Progress

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📬 Submit this assignment by **Monday, June 8th at 1:59PM GMT+7** using the *Submit* button 👉

Submit

Unit 1: Project 1 - Listicle Part 1

Overview

Who doesn't love a good list? In this project, you'll create a list-based web app that displays content of your choosing. Is it a guidebook to finding the best local music venues? 🎶 Is it a database of tips and tricks for young entrepreneurs? 💼 You'll create a web app that displays some interesting data in a list, building out a complete backend that serves static HTML, as well as a minimal frontend to serve the data.

👋 Keep in mind that in this project, you should **not** use a frontend framework, like React. Instead, use vanilla HTML/CSS/JS and Picocss to style your app.

> 💡 In Unit 2, you'll connect your app to a database. Make sure all items in your Listicle project have a few shared attributes. For example, in the Unearthed lab, each gift has a name, price point, description, and so on. See the examples below for more help.

Example App 1: Discover Local Music

Love live music? This app helps students find upcoming concerts, open mic nights, and music festivals happening near their college or university. Whether you're into indie, hip-hop, EDM, or acoustic sets, you can filter by genre, ticket price, or venue size. Users can click on an event to see more details, like the artist lineup, date and time, and ticket price.

Event shared attributes: event name, artist(s), date & time, venue, genre, ticket price (if applicable)

Example App 2: Guide to Starting Your Business

No matter what stage you are at opening up a business, we have a tutorial for you! Looking to brainstorm business ideas and find your niche? We can guide you in the right direction! Finding the right partner? Growing your team? Talking to suppliers? We have a short guide for each one of those. Our guides are sorted into different categories (including business models, product development, sales, market research, and funding) so you can browse or filter to personalize your experience.

Business tip shared attributes: title, text, category, image, submitted by

✨ **AI Opportunity**

▶ *Use AI to brainstorm ideas for your code*

🎯 Goals

By the end of this assignment you will be able to...

- [ ] Create a web server to handle incoming requests
- [ ] Create routes and request handlers using Express

Required Features

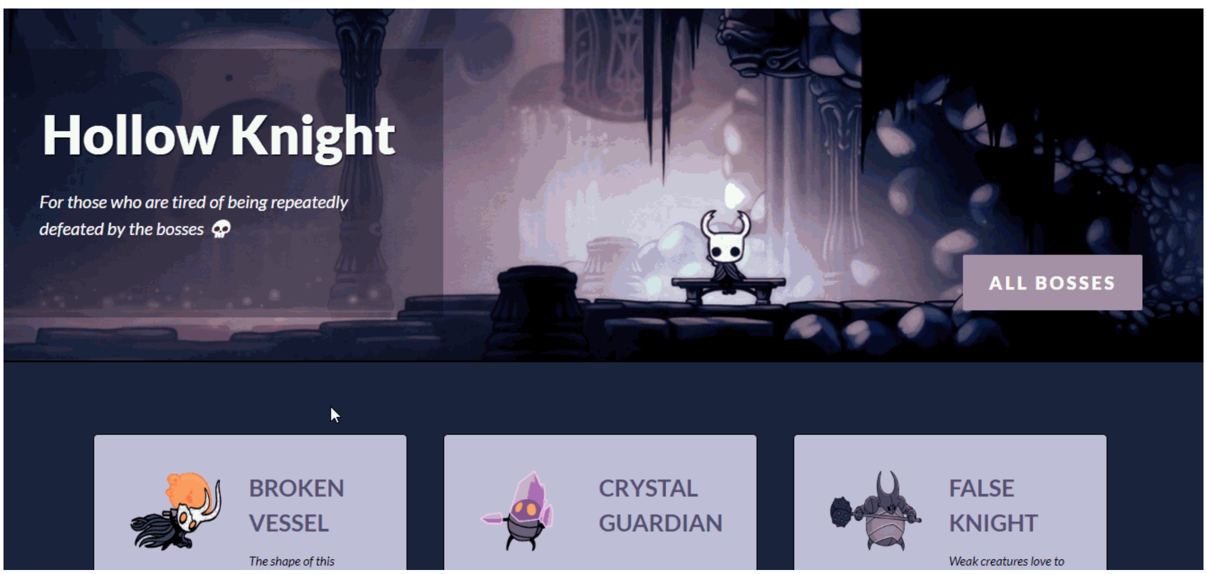
- [ ] The web app uses only HTML, CSS, and JavaScript without a frontend framework

- [ ] Front page of web app is functional and appropriately styled
    - [ ] The web app displays a title
    - [ ] Website displays at least five unique list items
    - [ ] Each list item includes at least three displayed attributes, such as a title, description, and image
- [ ] Each list item has a corresponding page
    - [ ] The user can click on each item in the list to see a detailed view of it, including all database fields
        - *e.g.,* `localhost:3000/bosses/crystalguardian` *and* `localhost:3000/mantislords`
    - [ ] The web app serves an appropriate 404 page when no matching route is defined
- [ ] The webpage is styled with Picocss

Screenshot

View an exemplar of the project [here](#)!

A simple version of the app with all the required features implemented:



Stretch Features

- [ ] List items are displayed in a unique format
    - *e.g., cards rather than lists or animated list items*

Resources

- [MDN Web Docs: HTML](#)
- [MDN Web Docs: CSS](#)
- [Picocss](#)

- [MDN Web Docs: JavaScript]
- [Express Routing]
- [Express Basic Routing]

💡 Hints

- Help! I don't know where to start!
  - After you decide on your topic, decide what information you want to display and how you want to display it on your web app. Then, figure out your routes and structure your pages.
  - Look at this week's lab for examples on how to implement similar applications. What code will be similar? What do you need to change?
- I'm stuck on something!
  - Don't just skip the Resources section.
  - Review the AI debugging tips in the box below!
  - Still need a little extra help getting started or running into an error? Try posting in the Class Slack Channel.

✨ AI Opportunity

▶ *Use AI to understand why your code isn't working*

📣 **Submission Guidelines**: For more details on submission instructions, follow the guidelines on the Submitting Coursework page.

- Make sure you are adding and committing files in git as you complete features and milestones.
- Be sure to **include a README** containing a GIF walkthrough of your project.
- Use this README TEMPLATE.
  - It is important that you follow the same layout as the README template so that we can easily access your work.
  - Be sure to check off each feature that is implemented in your submission by changing `[ ]` to `[x]`. We won't be able to assign points if a feature is unchecked.

Note: We highly encourage you submit your project in any state (even if it is not done) by **Monday, June 8th at 1:59PM GMT+7.** You can continue to work on your project with our *48-hour extension** in which your project will be graded once more once the extension deadline has passed (see "Coursework Submissions" in Syllabus for details). Don't forget to **resubmit through the course portal** with your **updated GIF recording**!